Fys4480/9480 AUG 21

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Ansatz for wf and expectation value of Energy.

$N = 2$ (Atomic helium)

$$\hat{H} = \hat{H}_0 + \hat{H}_I$$

$$\hat{H}_0 = \hat{h}_0(x_1) + \hat{h}_0(x_2) = \sum_{i=1}^{N=2} \hat{h}_0(x_i)$$

$$x_i = (\vec{r}_i, \sigma_i)$$

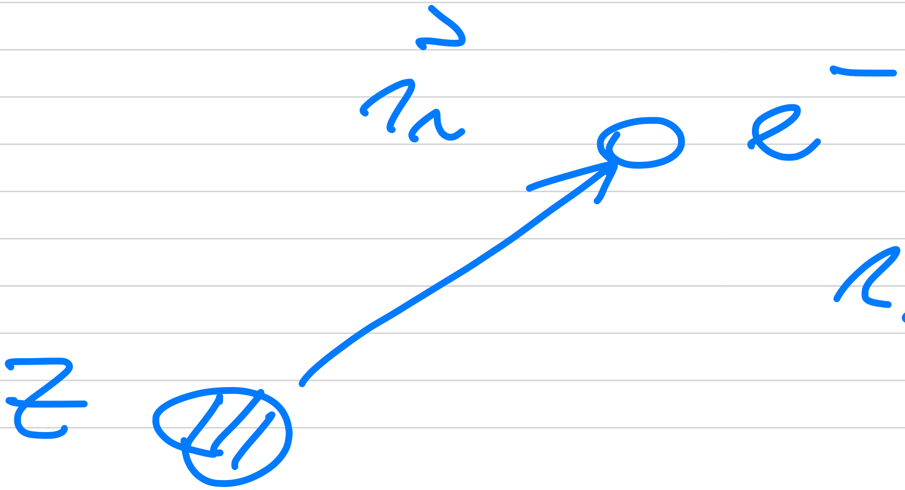
$\sigma_i = 1/2$ $\quad m_{\sigma_i} = \pm 1/2 \; (\uparrow \downarrow)$

$$\hat{h}_0(x_i)\,\varphi_\alpha(x_i) = \varepsilon_\alpha\,\varphi_\alpha(x)$$

Hydrogen-like orbitals

$$\hat{h}_0(x_i) = -\frac{\hbar^2}{2m}\nabla_r^2 - \frac{Z\cdot\gamma}{r_i}$$

$\vec{r}_i$ = distance from atomic nucleus and the electron



$$r_i = \sqrt{x_1^2 + y_1^2 + z_1^2}$$

$$\psi_\alpha(x) = \phi^{(r)}_{n l m_l} \otimes \xi_{\sigma m_\sigma}$$

$$\alpha = n\, l\, m_l\, \sigma\, m_\sigma$$

$$\begin{bmatrix} 1 \\ 0 \end{bmatrix} \text{ or } \begin{bmatrix} 0 \\ 1 \end{bmatrix}$$

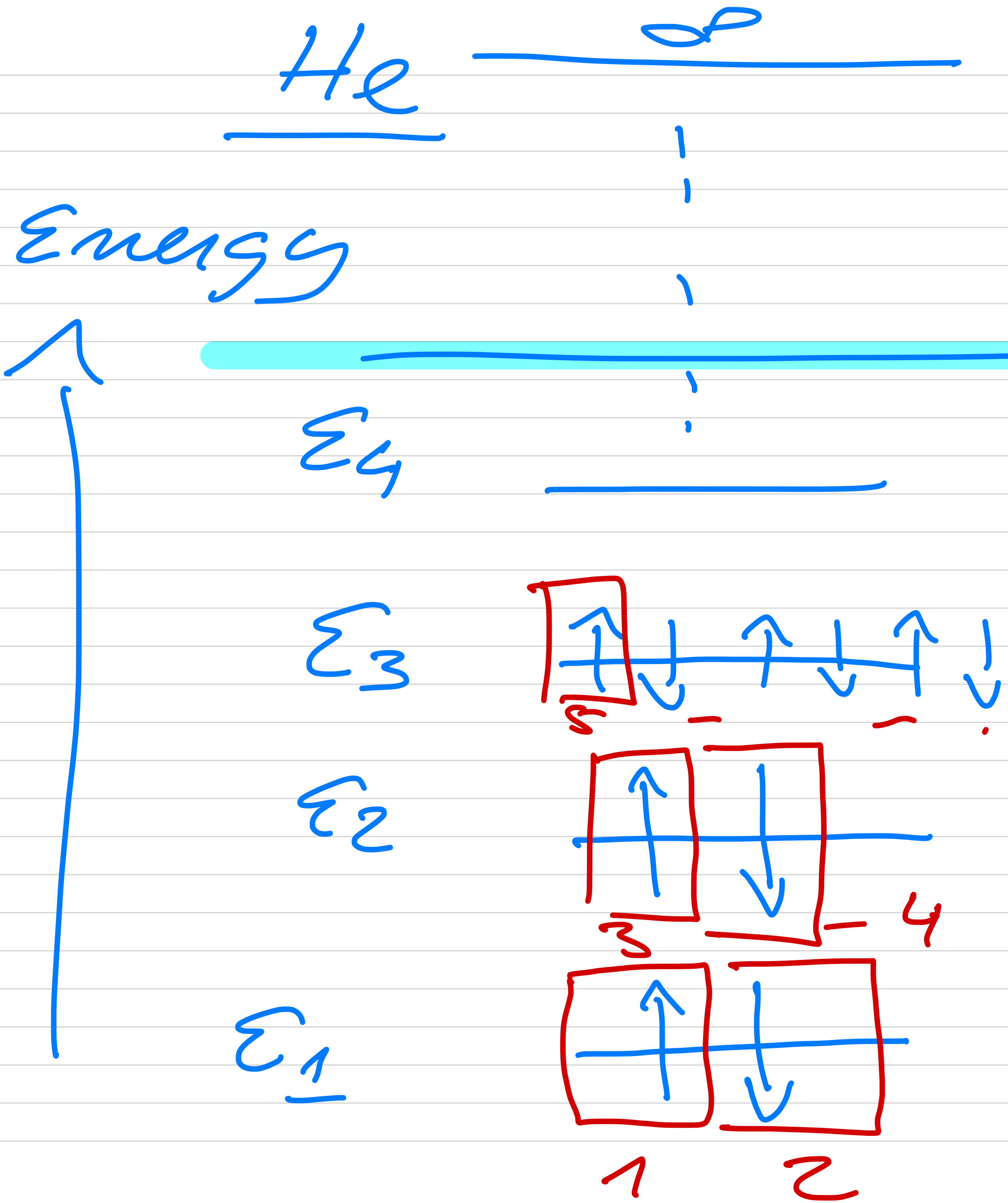
$m_\sigma = 1/2$

$m_\sigma = -1/2$

compact way

$$\psi_\alpha(x) \Rightarrow |\alpha\rangle$$

$$\psi_\alpha(x) : \phi^{(r)}_{n l m_l} = L_{nl}(r)\, Y_{l m_l}(\theta, \phi)$$
$$\times\, e^{-\beta r}$$



3s $n = 12$

2p
$n=2 \quad l=1 \quad m_l = -1, 0, +1$

2s
$n=2 \quad l=0 \quad m_l = 0$

1s
$n=1 \quad l=0 \quad m_l = 0$

$$\Phi_0(x_1, x_2) \propto \varphi_1(x_1)\,\varphi_2(x_2)$$

$$\hat{H}_0\,\Phi_0(x_1 x_2) = \mathcal{E}_0\,\Phi_0(x_1 x_2)$$

$$\Phi_0(x_1, x_2) = \frac{1}{\sqrt{2!}} \begin{vmatrix} \varphi_1(x_1) & \varphi_1(x_2) \\ \varphi_2(x_1) & \varphi_2(x_2) \end{vmatrix}$$

$$= \frac{1}{\sqrt{2!}}\left(\varphi_1(x_1)\,\varphi_2(x_2) - \varphi_1(x_2)\,\varphi_2(x_1) \right)$$

$$\mathcal{E}_0 = \varepsilon_1 + \varepsilon_2$$

states (configurations)

$$= \binom{n}{N} = \frac{12!}{(12-2)!\,2!}$$

$N = 2 \quad n = 12$

$$\frac{11 \cdot 12}{1 \cdot 2} = 66 \text{ states}$$

$$\langle \Phi_0 | \hat{H}_0 | \Phi_0 \rangle =$$

$$\frac{\int dx_1 \int dx_2 \; \Phi_0^*(x_1,x_2)\, \hat{H}_0 \, \Phi_0(x_1,x_2)}{\int dx_1 \int dx_2 \; \Phi_0^*(x_1,x_2)\, \Phi_0(x_1,x_2)}$$

$$\left(\int dx = \int d\vec{r} \sum_{\sigma m_\sigma} \right)$$

$$= \frac{1}{2} \int dx_1 \int dx_2 \left[\varphi_1^*(x_1)\varphi_2^*(x_2) - \varphi_1^*(x_2)\varphi_2^*(x_1) \right]$$

$$\left(\hat{h}_0(x_1) + \hat{h}_0(x_2) \right)$$

$$\times \left[\varphi_1(x_1)\varphi_2(x_2) - \varphi_1(x_2)\varphi_2(x_1) \right]$$

$$\langle \varphi_i | \varphi_j \rangle = \delta_{ij}$$

$$\hat{h}_0(x_i)\, \varphi_j^*(x_i) = \varepsilon_j\, \varphi_j(x_i)$$

(a)

$$\frac{1}{2} \int dx_1 \int dx_2\, \varphi_1^*(x_1)\, \varphi_2^*(x_2) \times \left(\hat{h}_0(x_1) + h_0(x_2) \right) \varphi_1(x_1)\, \varphi_2(x_2)$$

$$\underbrace{\frac{1}{2} \int dx_1\, \varphi_1^*(x_1)\, \hat{h}_0(x_1)\, \varphi_1(x_1) \int dx_2 \times \varphi_2^*(x_2)\, \varphi_2(x_2)}_{\varepsilon_1 + \varepsilon_2}$$

$$+ \langle h_0(x_2) \rangle$$

(ii) cross-term

$$-\frac{1}{2}\int dx_1 \int dx_2\, \varphi_1^*(x_1)\,\varphi_2^*(x_2)$$

$$\times\left(\hat{h}_0(x_1)+\hat{h}_0(x_2)\right)\varphi_1(x_2)\,\varphi_2(x_1)$$

$$\int dx_1 \quad \varphi_1^*(x_1)\,\hat{h}_0(x_1)\,\varphi_2(x_1)$$

$$\times \int dx_2\, \varphi_2^*(x_2)\,\varphi_1(x_2)$$

$$1 \neq 2 \qquad = 0$$

$$\langle \Phi_0 | \mathcal{H}_0' | \Phi_0 \rangle = \varepsilon_1 + \varepsilon_2$$

$$\varepsilon_i = \int dx \, \varphi_i^*(x) \, h_0(x) \, \varphi_i(x)$$

$$\langle \Phi_0 | \mathcal{H}_I | \Phi_0 \rangle$$

$$\frac{1}{2} \int dx_1 \int dx_2 \left(\varphi_1^*(x_1) \varphi_2^*(x_2) - \varphi_1^*(x_2) \varphi_2^*(x_1) \right)$$

$$\times \sum_{i<j} v(x_i, x_j)$$

$$\times \left(\varphi_1(x_1) \varphi_2(x_2) - \varphi_1(x_2) \varphi_2(x_1) \right)$$

$$v(x_i', x_j') = v(|x_i' - x_j'|)$$

$$v(x_i', x_j') = v(x_j', x_i')$$

(i)

$$\frac{1}{2}\int dx_1 \int dx_2\, \varphi_1^*(x_1)\, \varphi_2^*(x_2)\, v(x_1, x_2) \times \varphi_1(x_1)\, \varphi_2(x_2)$$

$$= \frac{1}{2} \langle 12 | v | 12 \rangle$$

(ii)

$$\frac{1}{2}\int dx_1 \int dx_2 \; \varphi_1^*(x_2)\,\varphi_2^*(x_1)\, v(x_1,x_2)\, \varphi_1(x_2)\,\varphi_2(x_1)$$

$$x_1 \longleftrightarrow x_2$$

$$v(x_1,x_2) = v(x_2,x_1)$$

$$= \frac{1}{2}\langle 21 | v | 21 \rangle = \frac{1}{2}\langle 12 | v | 12 \rangle$$

(iii)

$$-\frac{1}{2}\int dx_1 \int dx_2\, \varphi_1^*(x_1)\, \varphi_2^*(x_2)$$

$$v(x_1, x_2)\, \varphi_1(x_2)\, \varphi_2(x_1))$$

$$= -\frac{1}{2}\langle 12 | v | 21 \rangle$$

(iv)

$$-\frac{1}{2}\langle 21 | v | 12 \rangle$$

$$\langle \Phi_c | \hat{\mathcal{H}}_I | \Phi_c \rangle$$

$$= \underbrace{\langle 12 | v | 12 \rangle}_{\text{Direct}} - \underbrace{\langle 12 | v | 21 \rangle}_{\text{exchange}}$$

$$\langle \Phi_c | \hat{\mathcal{H}} | \Phi_c \rangle$$

$$= \varepsilon_1 + \varepsilon_2 + \langle 12 | v | 12 \rangle_{AS}$$

$$\langle ij|v|kl\rangle_{AS}$$

$$= \langle ij|v|kl\rangle - \langle ij|v|lk\rangle$$

$$\langle ij|v|kl\rangle = \langle ji|v|lk\rangle$$

$$\langle ij|v|kl\rangle_{AS} = \langle ji|v|lk\rangle_{AS}$$

$$= -\langle ij|v|lk\rangle_{AS}$$

$$= -\langle ji|v|kl\rangle_{AS}$$

with $N > 2$

$$\langle \Phi_0 | \hat{\mathcal{H}} | \Phi_0 \rangle = \underbrace{\sum_{i=1}^{N} \varepsilon_i}_{E_0} + \frac{1}{2} \sum_{ij}^{N} \langle ij | v | ij \rangle_{AS}$$